AI security engineering: review

AI security · University · Unit check



In this unit

- Threat-model an AI application
- The OWASP Top 10 for LLM applications
- Attacking and hardening models
- Red teaming and responsible disclosure

The big ideas

- Threat-model first: flows, boundaries, STRIDE, paths to actions
- OWASP: in, out, the system; model output is untrusted input
- Confidence isn't robustness; every defence has a cost
- Permission, a pass rate per category, the owner hears first

Design review

1. Teams of three get one made-up AI app
2. Five minutes: its top three risks and their controls
3. Swap: the other team tries to break your plan, on paper
4. Agree the fixes, and the test that proves each one

Everything stays on paper and made-up: no live systems.

Let’s discuss

1. Which practice will you use first on your own work?
2. Which question surprised you, and why?
3. What would you tell a classmate about testing systems without permission?

Check yourself

- My first practice from this unit, and the project I'll use it on